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# JAIPUR NATIONAL UNIVERSITY

## Semester/Year End Examination Dec.2024

|                             |                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Name of School:             | School of Engineering and Technology                                                                                                     |
| Name of Programme / Course: | B.Tech. (Computer Science) / B.Tech. CSE (Hons.) in Artificial Intelligence and Machine Learning / B.Tech. CSE (Hons.) in Cyber Security |
| Semester/Year:              | 3rd                                                                                                                                      |
| Paper Code:                 | BTECSCO303T22                                                                                                                            |
| Title of Paper:             | Digital Electronics                                                                                                                      |
| Time: 3 Hours               | Max. Marks: 60                                                                                                                           |

### INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section A and Section B. Section-A consists of seven questions and Section-B consists of six question. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

**Question  
Number**

**Question**

**Marks  
Allotted  
5X4=20**

#### Section A

~~Q.1~~ What are the Universal Gates & why they are called so?

~~Q.2~~ Represent Decimal Number (127) in Excess-3 and BCD Code?

~~Q.3~~ Describe Error Detecting & Correcting Codes?

~~Q.4~~ Explain HALF adder circuit also gives its circuit realization?

~~Q.5~~ Define Encoder with its truth table & Circuit Diagram?

~~Q.6~~ Differentiate between Synchronous and Asynchronous Counters?

~~Q.7~~ Design BCD to Excess-3 Codes.

## Section B

- Q.1 Explain Number System & Its Type with Suitable Examples?
- Q.2 Explain Multiplexers/De multiplexers & draw a 16:1 or 1:16 Logic Diagram?
- Q.3 Design of Synchronous FSM with suitable diagram?
- Q.4 Define Flip Flop & Draw R-S Flip With Circuit Diagram & its truth table?
- Q.5 How to work A/D Converter, Draw a circuit design?
- Q.6 Discuss Power Consumption & Gate Delay?